

ROUTES OF DRUG ADMINISTRATION





Most of the drugs can be administered by different routes.

Drug- and patient-related factors determine the selection of routes for drug administration.

• **The factors are:**

1. Characteristics of the drug.
2. Emergency/routine use.
3. Site of action of the drug—local or systemic.
4. Condition of the patient (unconscious, vomiting, diarrhea).
5. Age of the patient.
6. Effect of gastric pH, digestive enzymes, and first-pass metabolism.
7. Patient's/doctor's choice (sometimes).

Routes

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graph TD; Routes[Routes] --> Local[Local]; Routes --> Systemic[Systemic]; Systemic --> Enteral[Enteral]; Systemic --> Parenteral[Parenteral]; Enteral --> Oral[Oral]; Enteral --> sublingual[sublingual]; Enteral --> Rectal[Rectal]; Parenteral --> Inhalation[Inhalation]; Parenteral --> Injections[Injections]; Parenteral --> transdermal[transdermal];
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Local

Systemic

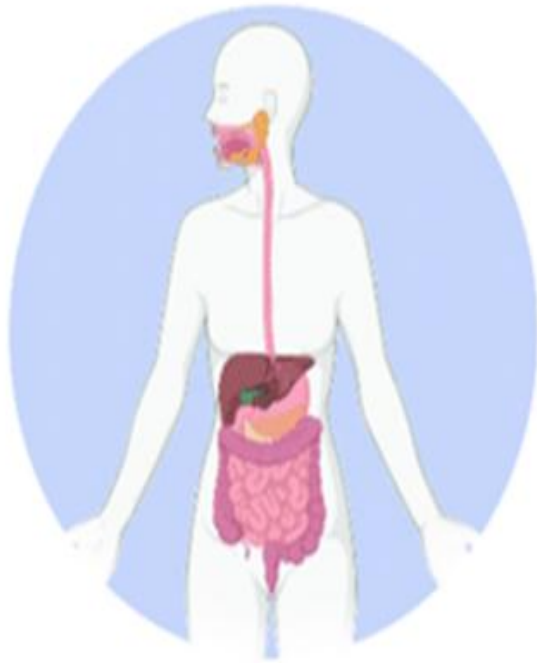
Enteral

- Oral
- sublingual
- Rectal

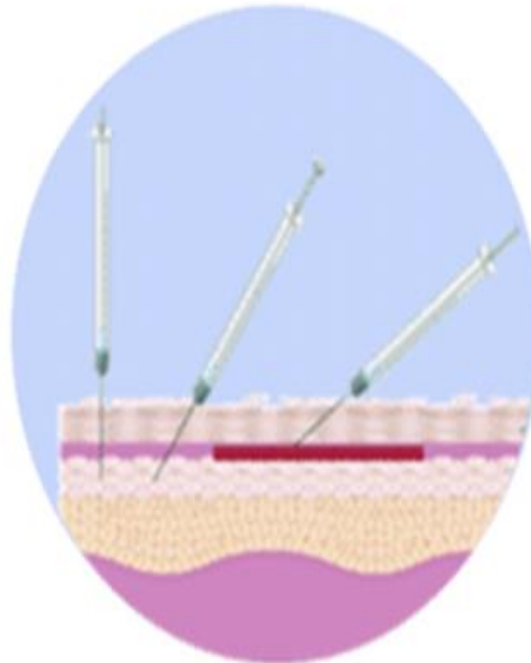
Parenteral

- Inhalation
- Injections
- transdermal

ROUTES OF DRUGS ADMINISTRATION



ENTERAL



PARENTERAL



TOPICAL



INHALATION

Local Routes

It is the simplest mode of administration of a drug at the site where the desired action is required.



Systemic side effects are minimal.

1. **Topical**: The drug is applied to the skin or mucous membrane at various sites for local action.

a. **Oral cavity**:

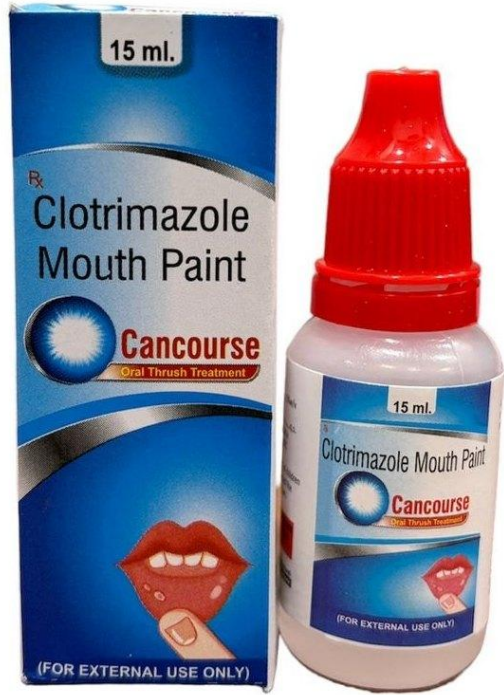
As a **suspension**, e.g., nystatin;

as a **troche**, e.g., clotrimazole (for oral candidiasis);

as a **cream**, e.g., acyclovir (for herpes labialis);

as **ointment** and **jelly**, e.g., 5% lignocaine hydrochloride (for topical anesthesia);

as a **spray**, e.g., 10% lignocaine hydrochloride (for topical anesthesia).





b. **GI tract:**

As a tablet that is not absorbed, e.g., **neomycin** (for sterilization of the gut before surgery).

c. Rectum and anal canal:

- i. As an **enema** (administration of a drug into the rectum in **liquid form**):
 - **Evacuant enema** (for evacuation of the bowel):
For example, soap water enema—soap acts as a lubricant and water stimulates the rectum.
 - **Retention enema**: **For example**, methylprednisolone in ulcerative colitis.
- ii. As a **suppository** (administration of the drug in a solid form into the rectum), **e.g.**, bisacodyl— for evacuation of bowels.

d. Eye, ear and nose:

As drops, ointments, and sprays (for infection, allergic conditions, etc.), **e.g.**, gentamicin eye/ear drops.

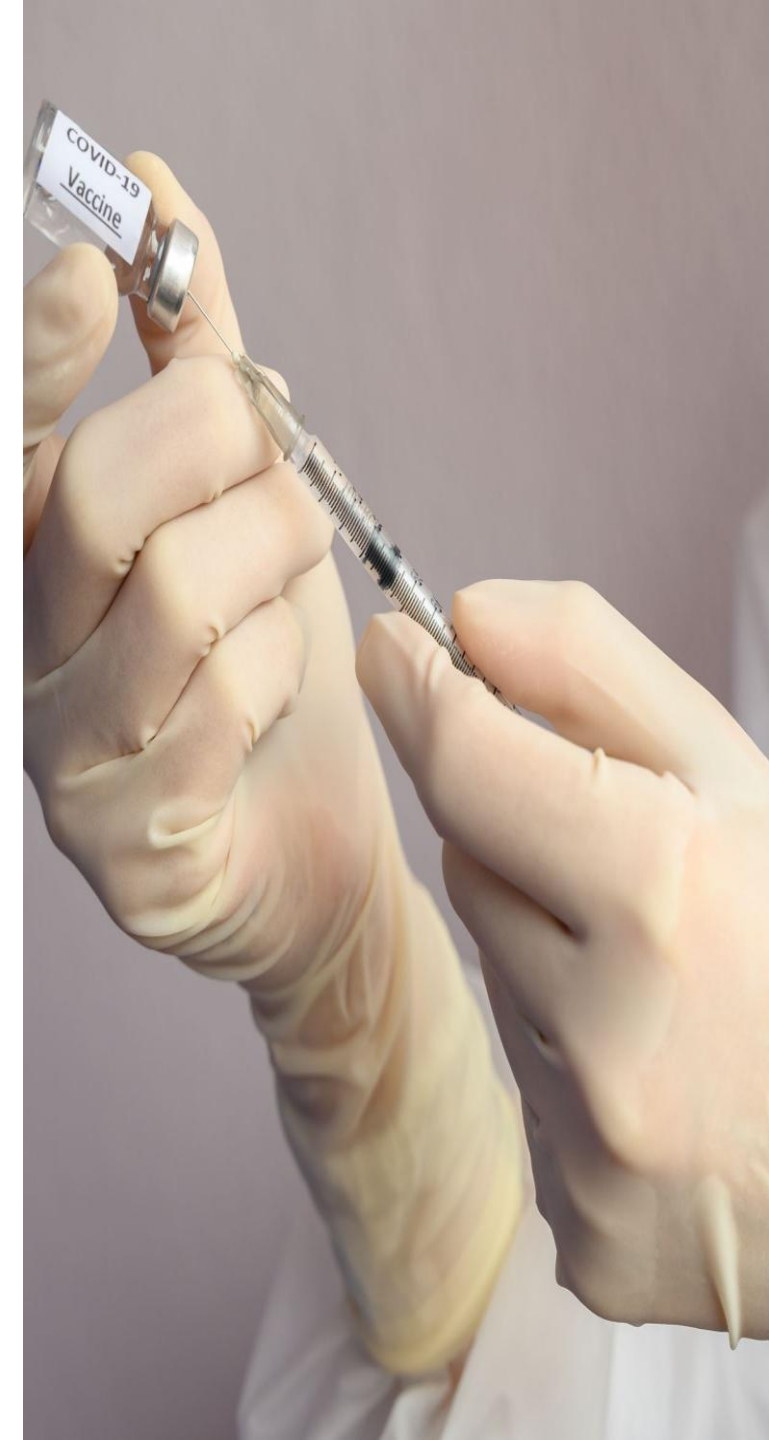
e. Bronchi: As an inhalation, **e.g.**, salbutamol, ipratropium bromide, etc. (for bronchial asthma and chronic obstructive pulmonary disease).

f. Skin: As ointment, cream, lotion, or powder, **e.g.**, clotrimazole (antifungal) for cutaneous candidiasis.

2. Intra-arterial route:

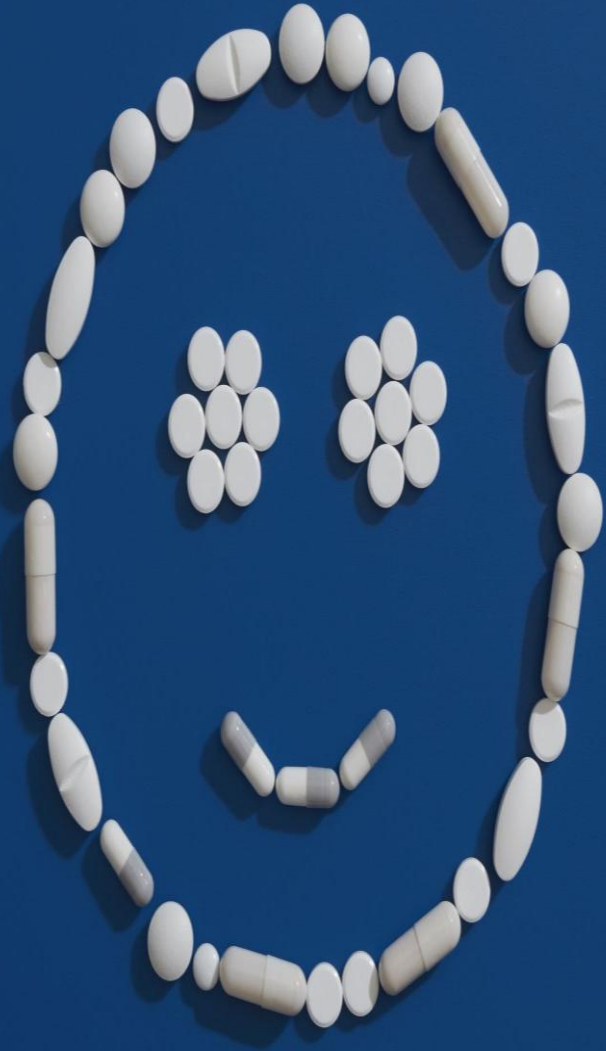
- This route is rarely employed. It is mainly used during diagnostic studies such as coronary angiography and for administering some anticancer drugs, **e.g.**, for treating malignancy involving limbs.

3. Administration of the drug into some deep tissues by injection, **e.g.**, administration of **triamcinolone** directly into the joint space in rheumatoid arthritis.



Systemic Routes

- Drugs administered by this route enter the blood and produce systemic effects.
- **Enteral Routes**
- It includes oral, sublingual, and rectal routes.
- **Oral Route**
- It is the most common and acceptable route for drug administration.
- Dosage forms are tablet, capsule, syrup, mixture, etc., **e.g.**, paracetamol tablet for fever, and omeprazole capsule for peptic ulcer are given orally.



- **Advantages**

- ✓ Safer.
- ✓ Cheaper.
- ✓ Painless.
- ✓ Convenient for repeated and prolonged use.
- ✓ Can be self-administered.

- **Disadvantages**

Not suitable for emergencies as the onset of action of orally administered drugs is slow.

It is not suitable for/in:

Unpalatable and highly irritant drugs.

Unabsorbable drugs (e.g., aminoglycosides).

Drugs that are destroyed by digestive juices (e.g., insulin).

Drugs with extensive first-pass metabolism (e.g., lignocaine).

Unconscious patients.

Uncooperative and unreliable patients.

Patients with severe vomiting and diarrhea.

Sublingual Route

- The preparation is kept under the tongue.
- The drug is absorbed through the buccal mucous membrane and enters the systemic circulation directly, **e.g.**, nitroglycerin for acute anginal attack and buprenorphine for myocardial infarction.

Advantages

- ✓ Quick onset of action.
- ✓ Action can be terminated by spitting out the tablet.
- ✓ Bypass first-pass metabolism.
- ✓ Self-administration is possible.

Disadvantages

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graph TD; A[Disadvantages] --> B[It is not suitable for:]; A --> C[Irritant and lipid-insoluble drugs.]; A --> D[Drugs with bad smell and taste.]
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**It is not
suitable for:**

Irritant and
lipid-insoluble
drugs.

Drugs with bad
smell and
taste.

Rectal Route

- **Drugs can be given in the form of solid or liquid.**

1. **Suppository**: It can be used for local (topical) effects as well as systemic effects, **e.g.**, indomethacin for rheumatoid arthritis.
2. **Enema**: Retention enema can be used for local effects as well as systemic effects.

The drug is absorbed through the rectal mucous membrane and produces a systemic effect, **e.g.**, diazepam for status epilepticus in children.

Parenteral Routes

- Routes of administration other than the enteral route are called parenteral routes.

- **Advantages**

The onset of action of drugs is faster; hence it is suitable for emergencies.

- **Useful in:**

- ✓ Unconscious patient.
- ✓ Uncooperative and unreliable patients.
- ✓ Patients with vomiting and diarrhea.

It is suitable for:

Irritant drugs.

Drugs with high first-pass metabolism

Drugs not absorbed orally.

Drugs destroyed by digestive juices.

Disadvantages

Require aseptic conditions.

Preparations should be sterile and expensive.

Requires painful, invasive techniques.

Cannot be usually self-administered.

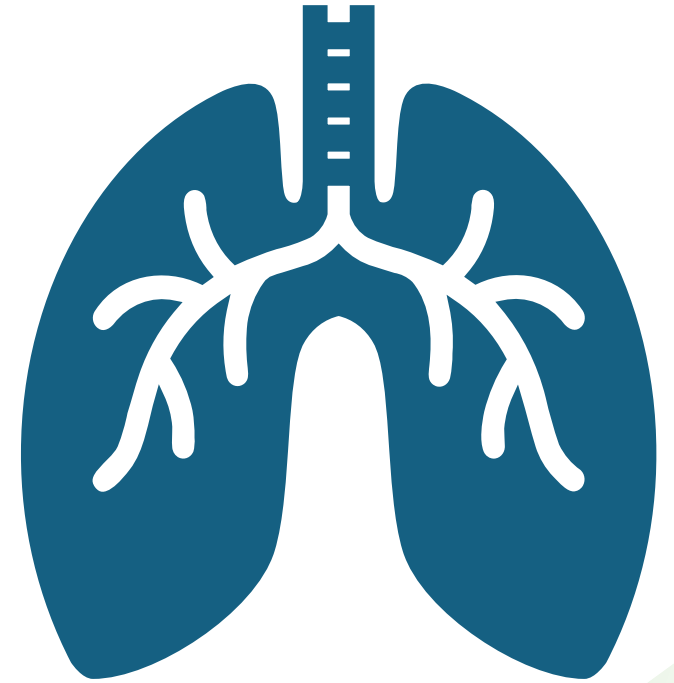
Can cause local tissue injury to nerves, vessels, etc.

Inhalation

- Volatile liquids and gases are given by inhalation for systemic effects, e.g., general anesthetics.
- **Advantages**
 - ✓ Quick onset of action.
 - ✓ Dose required is very less, so systemic toxicity is minimized.
 - ✓ Amount of drug administered can be regulated.

Disadvantages

Local irritation may cause increased respiratory secretions and bronchospasm



Injections

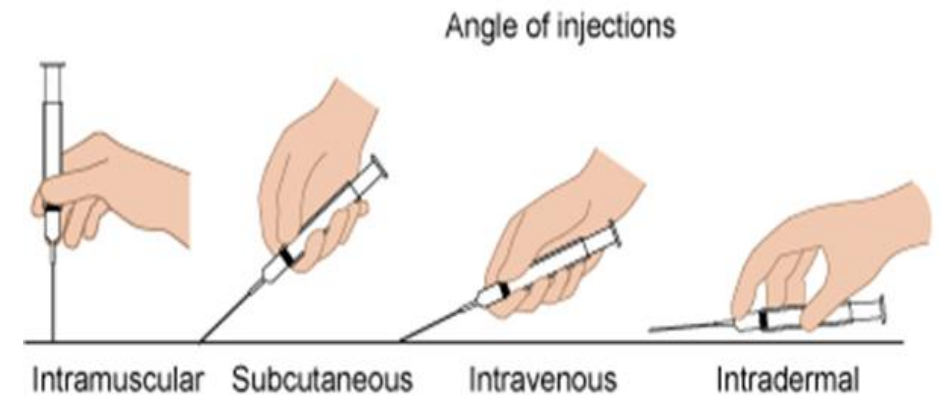
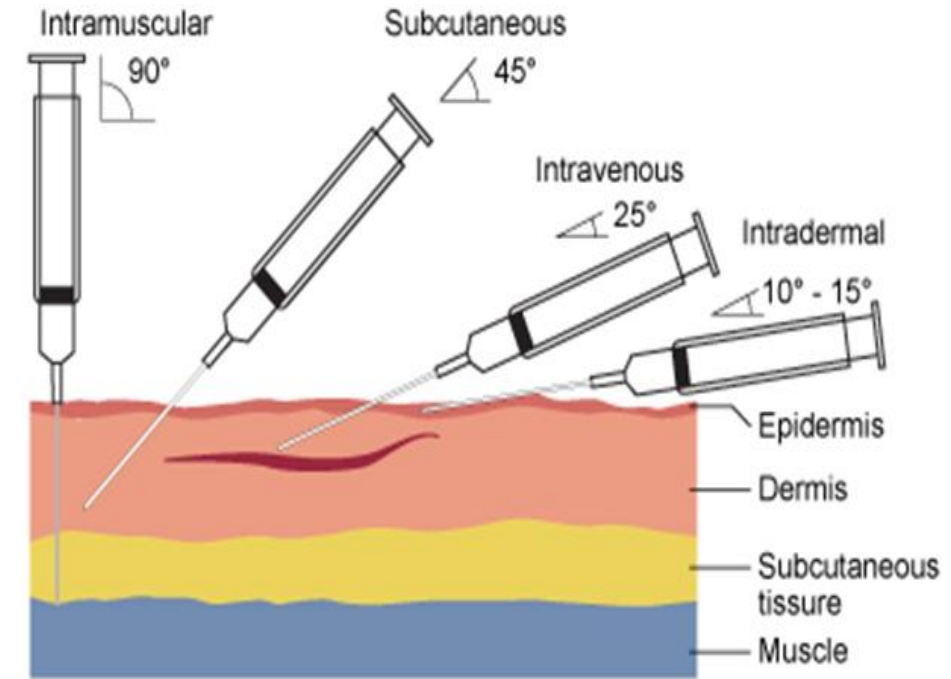
- **Intradermal route:**

The drug is injected into the layers of the skin, **e.g.** (BCG) vaccination and drug sensitivity tests.

It is painful and only a small amount of the drug can be administered.

- **Subcutaneous (s.c.) route:**

The drug is injected into the subcutaneous tissues of the thigh, abdomen, and arm, e.g. adrenaline, insulin, etc.



Advantages

Self-administration is possible (e.g., insulin).

Depot preparations can be inserted into the subcutaneous tissue, e.g., Norplant for contraception.

Disadvantages

It is suitable only for nonirritant drugs.

Drug absorption is slow; hence it is not suitable for emergencies.

- **Intramuscular (i.m.) route:** Drugs are injected into large muscles such as the deltoid, gluteus maximus, and vastus lateralis, **e.g.** paracetamol, diclofenac, etc.
- A volume of **5–10 mL** can be given at a time.
- **Advantages**
 - ✓ Absorption is more rapid as compared to the oral route.
 - ✓ Mild irritants, depot injections, soluble substances, and suspensions can be given by this route.
- **Disadvantages**
 - Aseptic conditions are needed.
 - Intramuscular injections are painful and may cause abscess.
 - Self-administration is not possible.
 - There may be an injury to the nerves

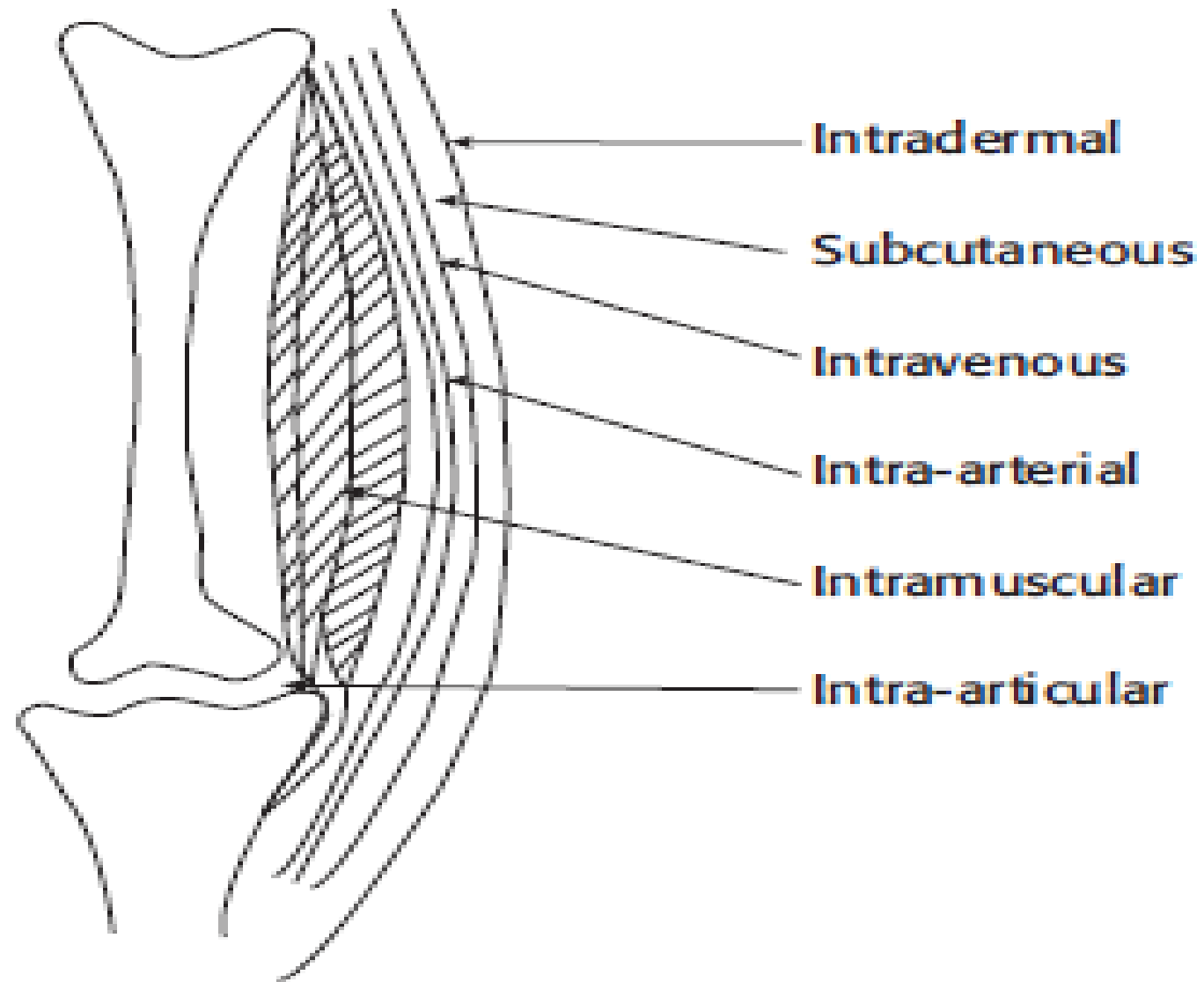
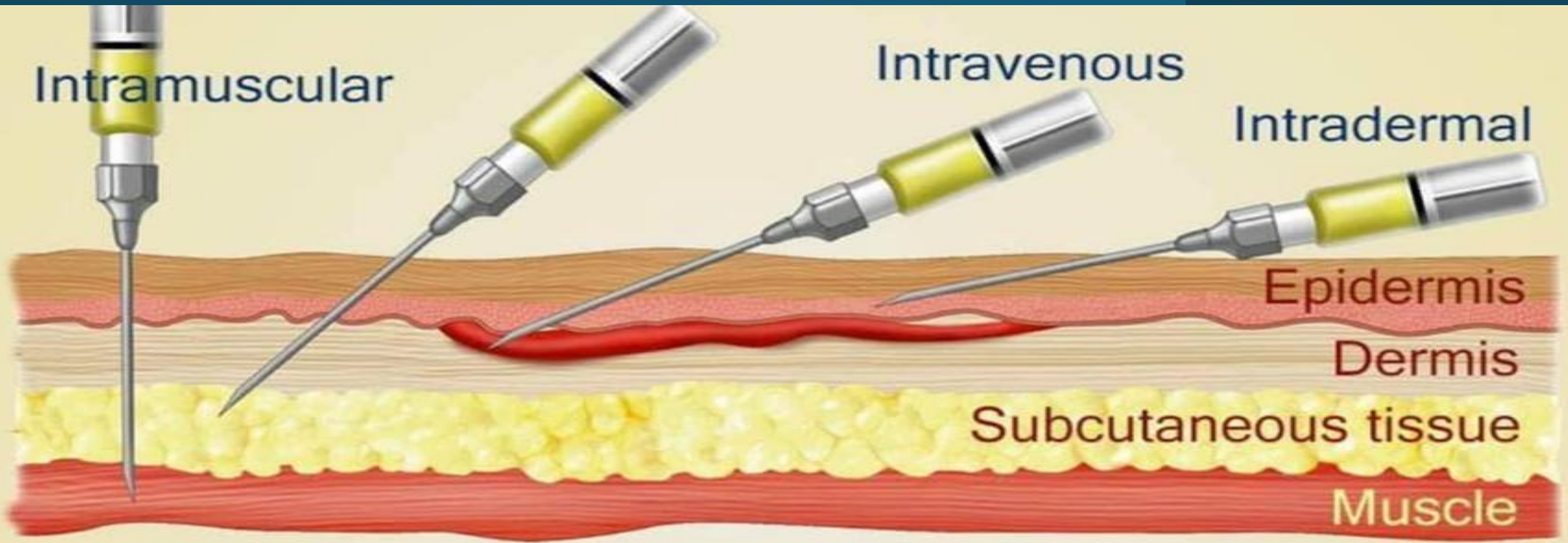


Fig. 1.1 Injectable routes of drug administration.

Intramuscular

Intravenous

Intradermal



Intramuscular

Subcutaneous

Intravenous

Intradermal



- **Intravenous (i.v.) route:**

- Drugs are injected directly into the bloodstream through a vein.

- **Drugs are administered as:**

1. **Bolus:** A single, relatively large dose of a drug injected rapidly or slowly as a single unit into a vein.

For example, i.v. ranitidine in bleeding peptic ulcer.

2. **Slow intravenous injection:**

For example, i.v. morphine in myocardial infarction.

3. **Intravenous infusion:**

For example, dopamine infusion in cardiogenic shock; mannitol infusion in cerebral edema; and fluids infused intravenously in dehydration.

Advantages

Bioavailability is 100%.

Quick onset of action; therefore, it is the route of choice in emergency, **e.g.**, intravenous diazepam to control convulsions in status epilepticus.

Large volume of fluid can be administered, **e.g.**, intravenous fluids in patients with severe dehydration.

Highly irritant drugs, **e.g.**, anticancer drugs can be given because they get diluted in blood.

Hypertonic solution can be infused by the intravenous route, **e.g.**, 20% mannitol in cerebral edema.

By i.v. infusion, a constant plasma level of the drug can be maintained, **e.g.**, dopamine infusion in cardiogenic shock.

Disadvantages

Once the drug is injected, its action cannot be halted.

Local irritation may cause phlebitis.

Self-medication is not possible.

Strict aseptic conditions are needed.

Extravasation of some drugs can cause injury, necrosis, and sloughing of tissues.

Depot preparations cannot be given by i.v. route.

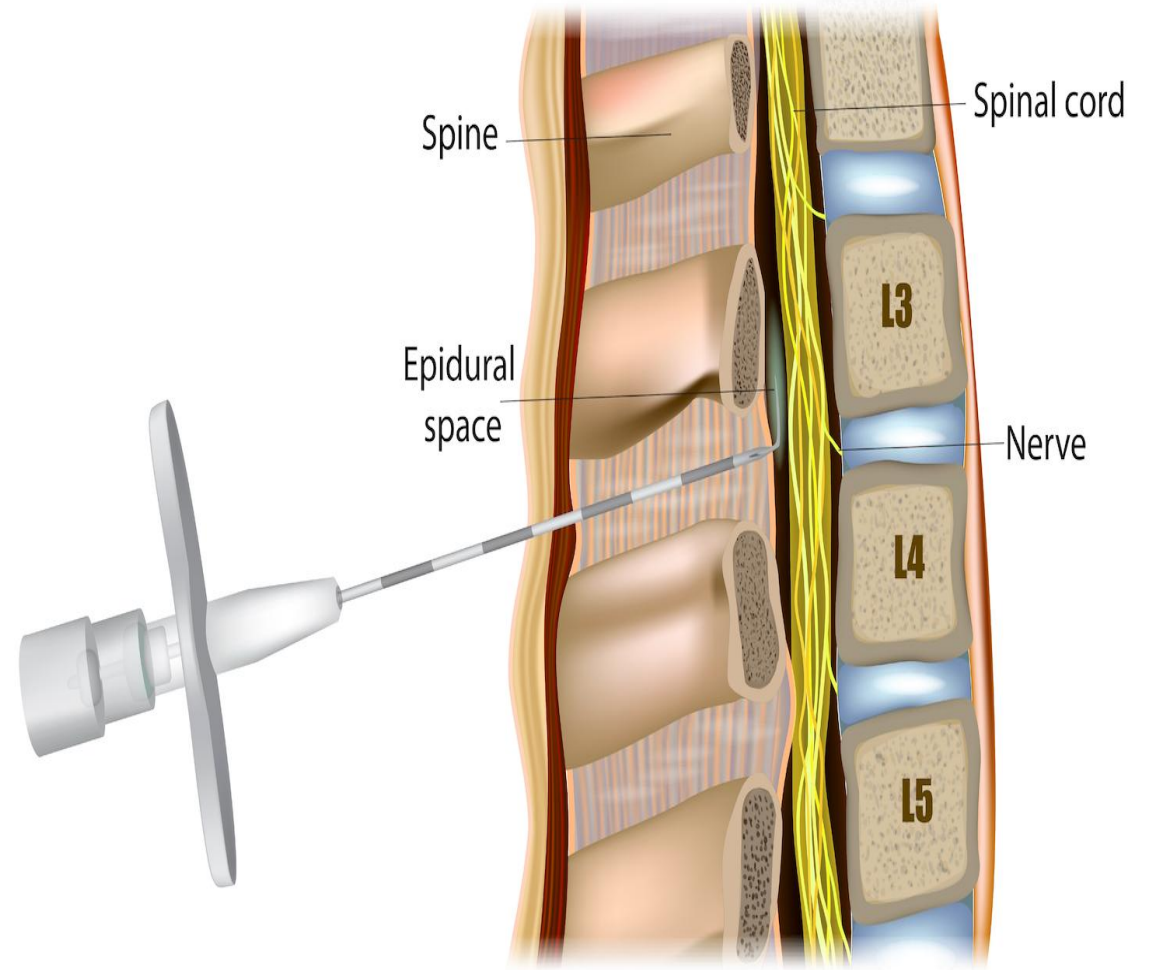
Precautions

Drug should usually be injected slowly.

Before injecting, make sure that the tip of the needle is in the vein.

• Intrathecal route:

- Drug is injected into the subarachnoid space (spinal anesthetics)



Intra-articular route

- Drug is injected directly into the joint space, **e.g.**, hydrocortisone injection for rheumatoid arthritis.
- Strict aseptic precautions should be taken.
- Repeated administration may cause damage to the articular cartilage.

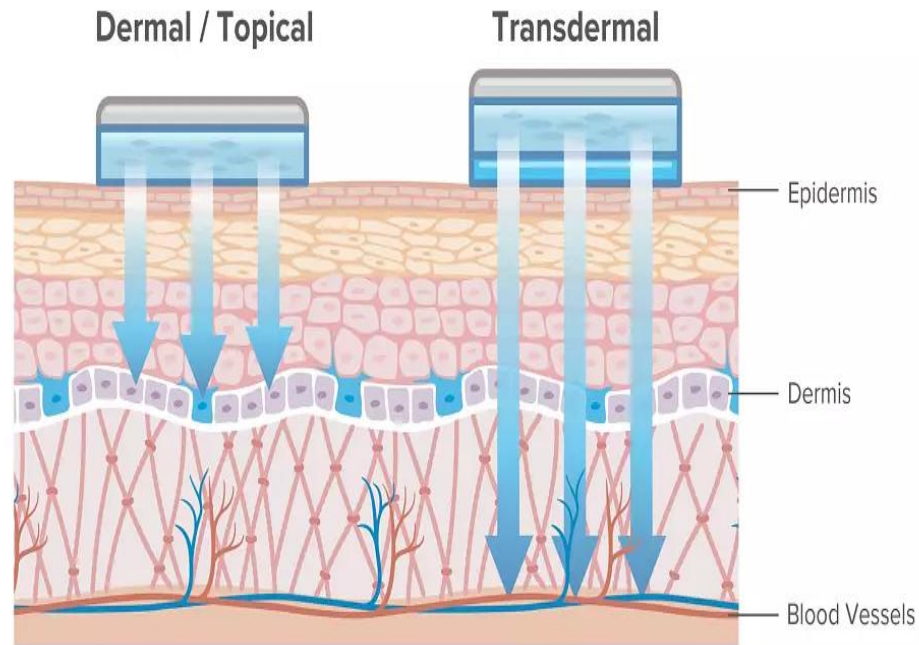
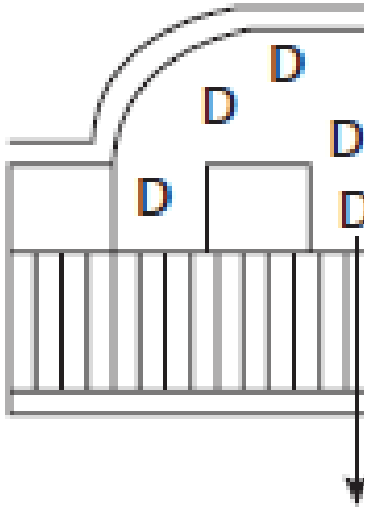


- **Transdermal route:**

The drug is administered in the form of a patch or ointment that delivers the drug into the circulation for systemic effect

- **For example**, nitroglycerin patch for angina





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- **Advantages**

- ✓ Self-administration is possible.
- ✓ Patient compliance is better.
- ✓ Duration of action is prolonged.
- ✓ Systemic side effects are reduced.
- ✓ Provides a constant plasma concentration of the drug.

- **Disadvantages**

- Expensive.
- Local irritation may cause dermatitis and itching.
- Patch may fall off unnoticed.

Special Drug-Delivery System

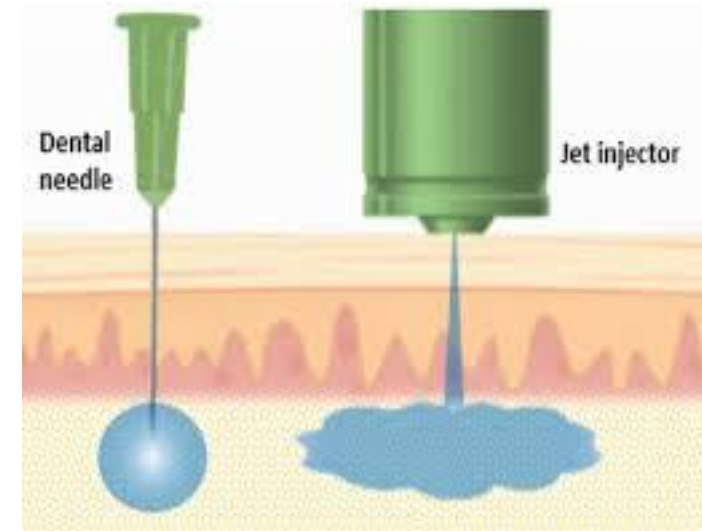
1. Intraoral lignocaine patch:



Patch containing lignocaine is used to anesthetize the oral mucosa.

2. Jet injection:

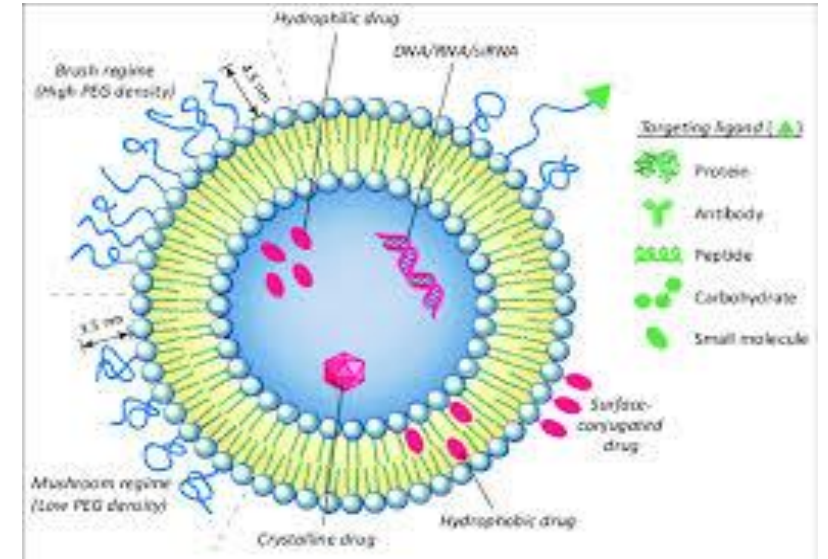
Small amount of local anesthetic can be administered into the submucosa without the use of a needle to produce surface anesthesia.



3. Liposomes:

They are minute vesicles made of phospholipids into which the drug is incorporated.

They help in targeted delivery of drugs, **e.g.**, liposomal formulations of amphotericin B for fungal infections.



5. Monoclonal antibodies:

They are **immunoglobulins**, produced by cell culture, selected to react with a specific antigen.

They are useful for targeted delivery of drugs, **e.g.**, delivery of anticancer drugs using monoclonal antibodies.

